Engineering Mechanics
2 Marks

UNIT I – FUNDAMENTAL OF MECHANICS

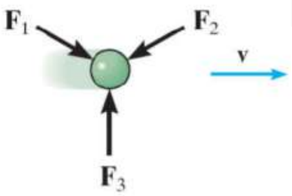
1. **Define force.**

   Force is a physical quantity that changes or tires to change the state of rest or of uniform motion of an object. Unit of force is newton 'N'.

2. **Differentiate between particles and rigid body.**
   i. Particle is a body, which has mass but no dimension whereas rigid body as both mass and dimensions.
   ii. Particle can have only translational motion whereas rigid body can have translational as well as rotational motion.

3. **State Newton's first law of motion.**

   Everybody tries to be in it's state of rest or of uniform motion along a straight line unless it's acted upon by an external unbalanced force.



4. **State Newton's second law of motion.**

   The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

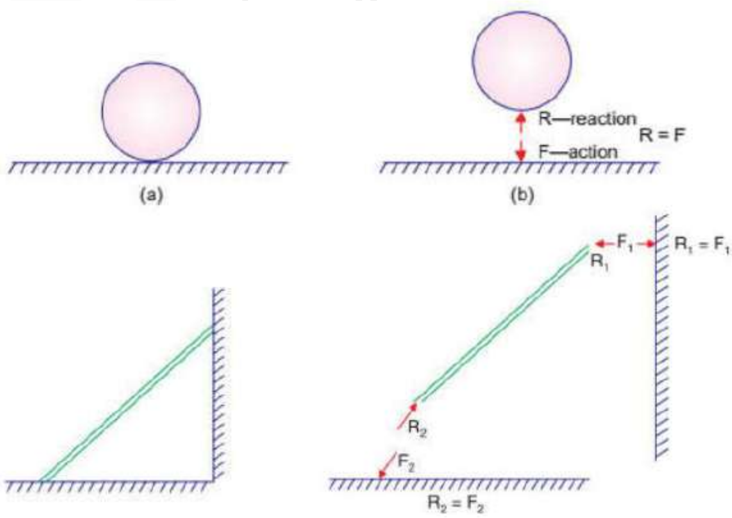
$$F=ma$$

Where

F-Force (N)
m-Mass (kg)
a – acceleration (m/sec^2)

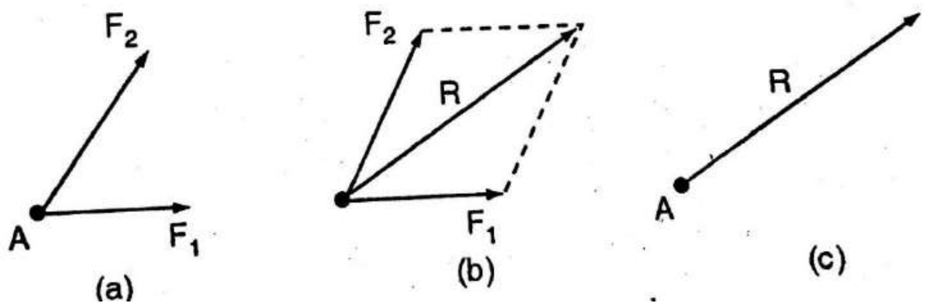
5. **State Newton's third law of motion.**

   For the Every action there has an equal and opposite reaction.



6. **State law of parallelogram of vectors.**

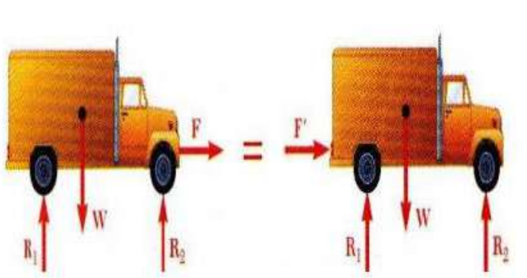
   If two vectors are represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant is represented in magnitude and direction by the diagonal of the parallelogram drawn from the common point.

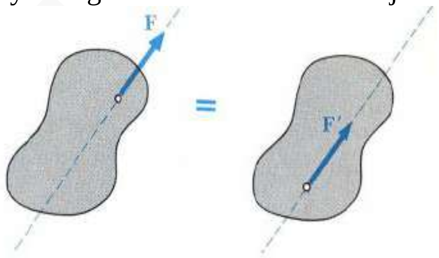


$$R = \sqrt{F_1^2 + F_1^2 + 2F_1F_2cas\theta}$$

7. **State the principle of transmissibility of force with simple sketch.**

According to principle of transmissibility of force, the force can be transmitted from one point to another on it is line of action without causing by any change in the motion of the object.



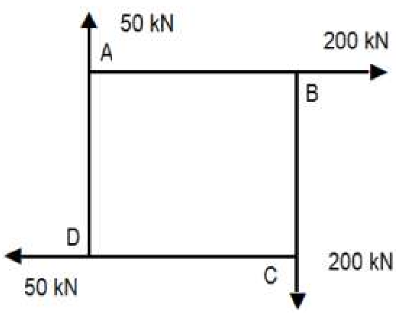


8. **Define unit vector.**

A vector having magnitude one unit is known as a unit vector .

9. **Define the following terms. (a)coplanar forces, (b)concurrent forces.**

a) Coplanar Force Systems: When the system of forces are in a plane, it is called coplanar system of forces.

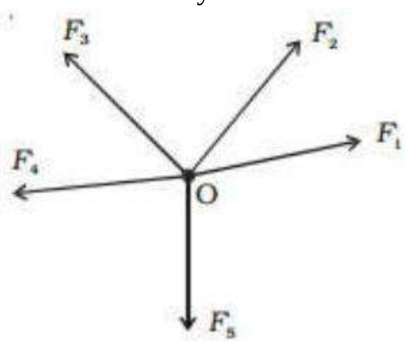


Coplanar force

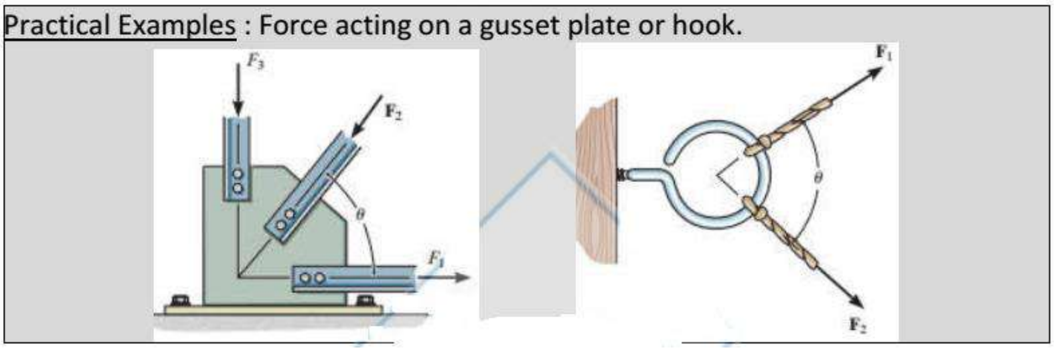
b) Concurrent Force System: A concurrent force system contains forces whose lines-of action meet at same one point.

i) Coplanar Concurrent System

ii) Non- Coplanar Concurrent System



Coplanar Concurrent System

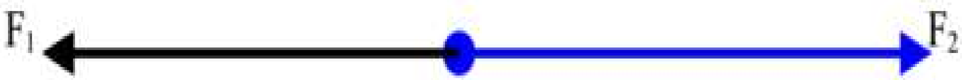


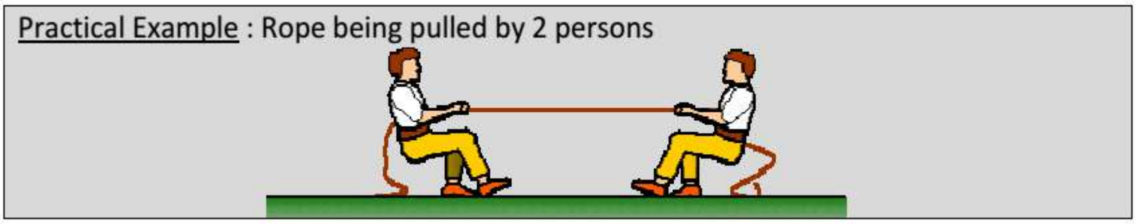
Coplanar Concurrent System

10. **Differentiate between collinear and concurrent forces.**

<u>Collinear Force System :</u>

When a system of forces act along the same line, they are called collinear forces.

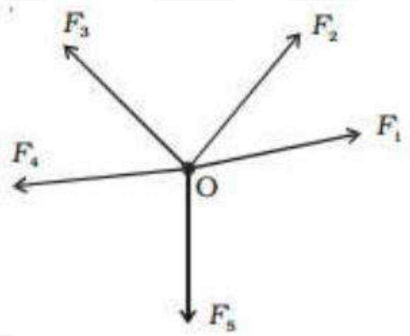




<u>**Concurrent Force System:**</u>

A concurrent force system contains forces whose lines-of action meet at same one point.

iii) Coplanar Concurrent System
iv) Non- Coplanar Concurrent System



Coplanar Concurrent System

11. **Define resultant of coplanar concurrent force system.**

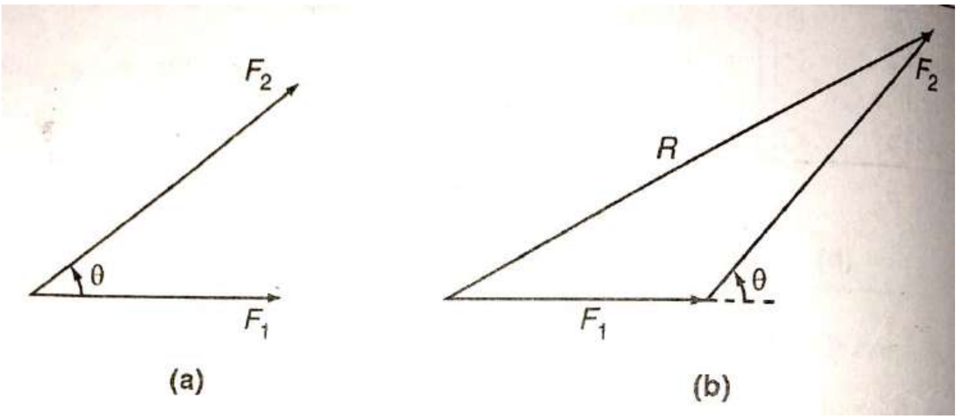
A system of coplanar concurrent forces can be reduced to a single force, which is known as resultant force.

12. **What is the difference between a resultant force and equilibrant force?**

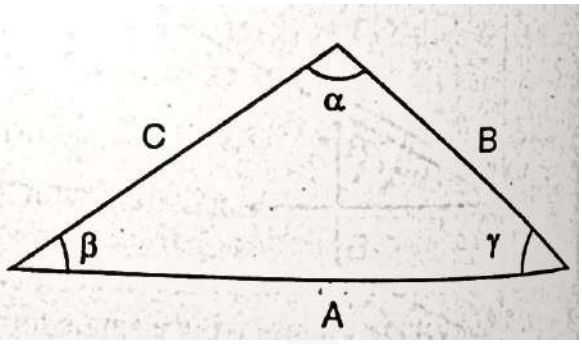
Resultant force makes the object move whereas equilibrant force keeps it in equilibrium.

13. **State triangle law of forces.**

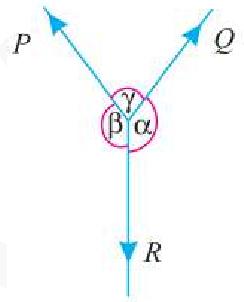
It states, "If two forces acting simultaneously on a particle, be represented in magnitude and direction by the two sides of a triangle, taken in order ; their resultant may be represented in magnitude and direction by the third side of the triangle, taken in opposite order."



$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}$$
$$A^2 = B^2 + C^2 - 2BC\cos\alpha$$
$$B^2 = A^2 + C^2 - 2AC\cos\beta$$
$$C^2 = A^2 + B^2 - 2AB\cos\gamma$$



14. **State lami's theorem.**
It states, "If three coplanar forces acting at a point be in equilibrium, then each force is proportional to the sine of the angle between the other two."

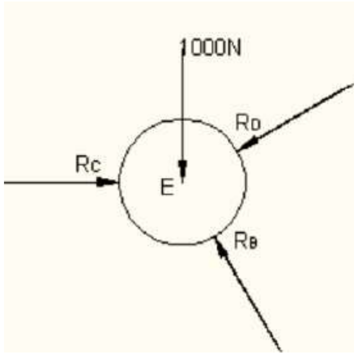


$$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$$

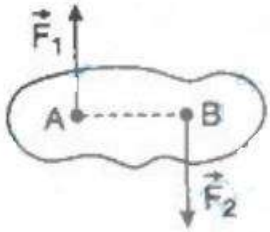
15. **State the necessary and sufficient condition for static equilibrium of a particle in two dimensions.**
For static equilibrium of a particle in two dimensions,
    i. The algebraic sum of horizontal components of all forces acting the particle must be zero.
    ii. The algebraic sum of vertical components of all forces acting the particle must be zero.
    iii. Algebraic sum of moments due to all forces and couple moments acting the on the bodies must be zero.

$$\sum F_x = 0, \sum F_y = 0, \sum M = 0$$

16. **What is a free body diagram?**
It is the sketch of the isolated body which shows the external forces on the body and the reaction exerted on it by the removed elements.



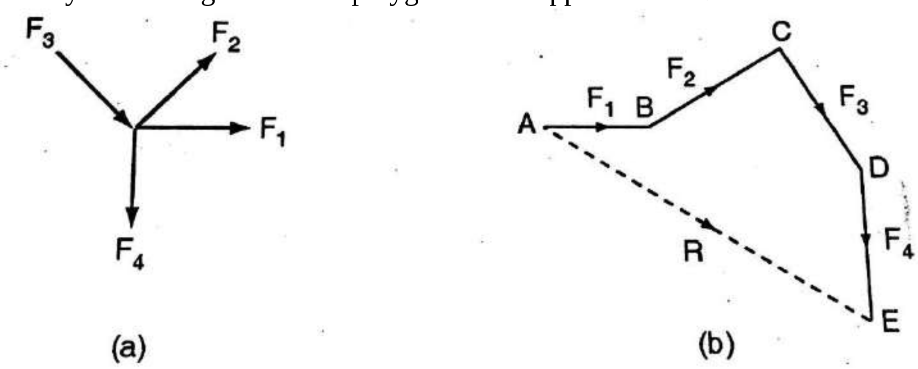
17. **What is a rigid body**
A body whose deflection when subjected to loading is negligible when compared to the dimensions of the body is called as a rigid body.**(body with negligible deformation).**
18. **What is a couple?**
The turning effect produced by two equal and opposite force separated by a distance constitute a couple.



19. **What is moment of force?**
It is the turning effect produced by a force, on the body, on which it acts. The moment of a force is equal to the product of the force and the perpendicular distance of the point, about which the moment is required and the line of action of the force.
$M = P \times l$
where P = Force acting on the body, and
l = Perpendicular distance between the point, about which the moment is required and the line of action of the force.
20. **What is equilibrant force?**
Equilibrant force is the force which is opposite to the resultant force and it brings the system to equilibrium.
21. **State varignon's theorem.**
If a number of coplanar forces are acting simultaneously on a body, the sum of moments of all the forces about any point is equal to the moment of the resultant force about the same point.

$$\sum m = \sum R \times X$$

22. **State Polygon Law of force.**
If a number of concurrent forces acting simultaneously on a particle, are represented in magnitude and direction by the sides of the polygon taken in order, then the resultant of this system of forces is represented by the closing side of the polygon on the opposite order.

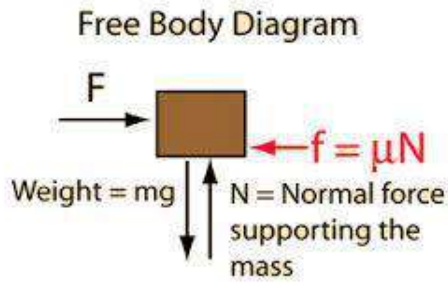


23. **What is you mean by resolution of forces?**

The process of replacing a single force F acting on a particle by two or more forces which together have the same effect as that of a single fore is called resolution of forces.

UNIT II – PRACTICAL APPLICATION OF FORCE SYSTEM

1. **What is a free body diagram?**
A free-body diagram or isolated-body diagram is useful in problems involving equilibrium of forces. Free-body diagrams are useful for setting up standard mechanics problems.



2. **Define couple.**
Torque is calculated by the product of either of forces forming the couple and the arm of the couple. i.e.

Torque = (one of the force) x (perpendicular distance between the forces.)

3. **Why the couple moment is said to be a free vector?**
   - Couple moment is said to be a free vector as it can be transferred to any point in the plane without causing any change in its effect on the body.
   - In mechanics, a couple is a system of forces with a resultant moment but no resultant force.
   - A better term is force couple or pure moment. Its effect is to create rotation without translation, or more generally without any acceleration of the centre of mass. In rigid body mechanics, force couples are free vectors, meaning their effects on a body are independent of the point of application.

4. **Distinguish between couple and moment.**
   - Moment of a force about a point is the measure of its rotational effect. Moment is defined as the product of the magnitude of the force and the perpendicular distance of the point from the line of action of the force from that point.
   - Two parallel forces equal in magnitude and opposite in direction and separated by a definite distance are said to form a couple. The sum of the forces forming a couple is zero, since they are equal and opposite which means the translatory effect of the couple is zero,

5. **What is meant by force-couple system?**
A system of coplanar non concurrent force system acting in a rigid body can be replaced by a single resultant force and couple moment at a point known as force couple system. Two force systems are equivalent if they result in the same resultant force and the same resultant moment.

6. **When is moment of force zero about a point?**
A force produces zero moment about an axis or reference point which intersects the line of action of the force.

7. **What are the three equations of equilibrium?**
The condition of a system in which all competing influences are balanced, in a wide variety of contexts.

$\Sigma Fx=0,$
$\Sigma Fy=0,$
$\Sigma M=0.$

8. **What are the characteristics of a couple?**
There are two types of characteristics of a couple
i) The algebraic sum of the forces is zero
ii) The algebraic sum of the moments of the forces about any point is the same and equal to the moment of the couple itself.

9. **List out the types of supports.**
a) Hinge support. (Or) pin support.
b) Roller support.
c) Fixed support.
d) Double overhanging
e) Continuous
f) Trussed

10. **List out the types of beams.**
a) Cantilever beam.
b) Fixed beam.
c) Simply supported beam and
d) Over hanging beam

11. **Write short notes on Resultant.**
Definition: A resultant of number of forces acting on a body is a single. Force which can produce the same effect on the body as it is produced by all the forces acting together. Resultant (net) force causes the displacement of a body (i.e. body moves). The set of forces which causes the displacement of a body are called as component of resultant or component forces.

12. **Write short notes on equilibrant.**
Definition: An equilibrant of number of forces acting on a body is a single force which cancels the effect of resultant of a system of forces or which brings the system and the body is equilibrium. Equilibrant keeps the body at rest (i.e. in equilibrium). The set of forces which keeps the body at restate known as equilibrium forces or components of equilibrant.

13. **Write short notes about Moment of a Force about an Axis.**
Two forces of the same size and direction acting at different points *are not equivalent.* They may cause the same translation, but they cause different rotation. For straight members, moments (torques) are explicitly excluded because, and only because, all the joints in a truss are treated as revolute, as are necessary for the links to be two-force members.

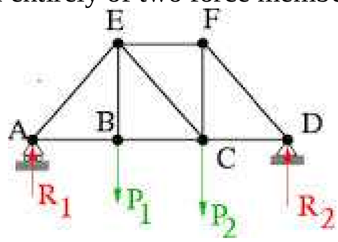
14. **What are the types of loads?**
There are three types of loads such as,
a) Point load (OR) concentrated load,
b) Uniform distributed load, and
c) Uniform varying load.

15. **What is stable equilibrium?**

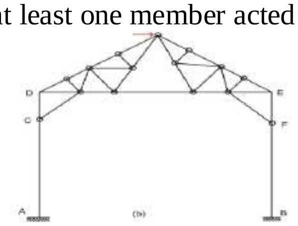
A body is said to be in stable equilibrium, if it returns back to its original position after it is slightly displaced from its position of rest.

16. Write short notes on trusses and frames.

Trusses: Structures composed entirely of two force members.(tensile & compressive)



Frames: Structures containing at least one member acted on by forces at three or more points.



17. What is statically determinate structure?

A structure which can be completely analysed by static conditions of equilibrium ($\sum H = 0$; $\sum V = 0$ and $\sum M = 0$) alone is statically determinate structure.

18. What is statically indeterminate structure?

A structure which **cannot** be analysed by static conditions of equilibrium ($\Sigma H = 0$; $\Sigma V = 0$ and $\Sigma M = 0$) alone is statically indeterminate structure.

19. Define friction.

Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. A resistance encountered when one body moves relative to another body with which it is in contact. The property of body by virtue which a force is exerted by a stationary body on the moving body to resist the motion of the moving body is called friction.

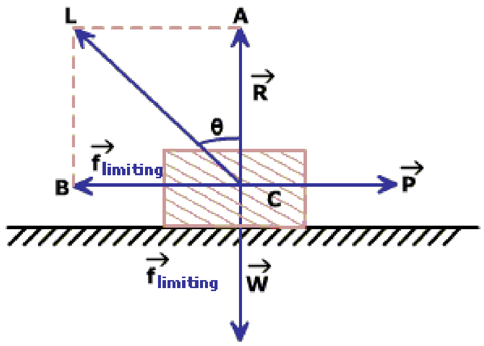
20. Define static and dynamic or kinetic friction.

Static friction is friction between two or more solid objects that are not moving relative to each other. For example, static friction can prevent an object from sliding down a sloped surface. The coefficient of static friction, typically denoted as μs, is usually higher than the coefficient of kinetic friction. The static friction force must be overcome by an applied force before an object can move. The maximum possible friction force between two surfaces before sliding begins is the product of the coefficient of static friction and the normal force.

Kinetic friction, or dynamic friction, is defined as the form of friction between two bodies via their surface of contact (i.e., the force acting parallel to the plane of contact) when the two surfaces of contact are slipping against each other. Kinetic friction is the form of friction that opposes the slipping against each other of two surfaces in contact. Its direction is opposite the direction of relative motion of the surfaces in contact. Note that it need not oppose the net external force between the surfaces in contact. Kinetic friction refers to the frictional force of a moving object.

21. What is angle of friction?

Consider a block placed on a rough floor. Now, the reaction force is because it is equal and opposite to the weight. Now apply a horizontal force so that the block just begins to slide, i.e., the frictional force is equal to the limiting friction. When this condition is satisfied, the angle which the resultant (between the normal and external force) makes with the vertical is called the angle of friction.

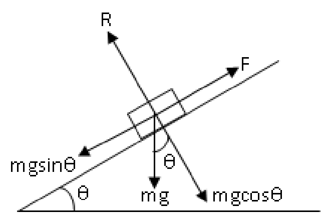


Angle of friction

22. Define coefficient of static and dynamic friction.

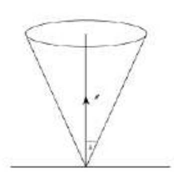
A coefficient of friction is a value that shows the relationship between the force of friction between two objects and the normal force between the objects. It is a value that is used in physics sometimes to find an objects normal force or frictional force, when other methods aren't available. The coefficient of friction is dimensionless, meaning it does not have any units. It is a scalar, meaning the direction of the force does not change its magnitude.

23. What is angle of repose?



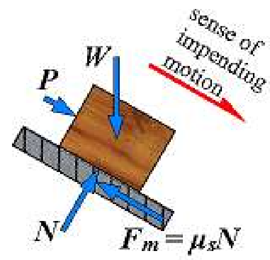
Angle of repose is defined as the minimum angle made by an inclined plane with the horizontal such that an object placed on the inclined surface just begins to slide.

24. Define cone of friction.



A cone in which the resultant force exerted by one flat horizontal surface on another must be located when both surfaces are at rest, as determined by the coefficient of static friction.

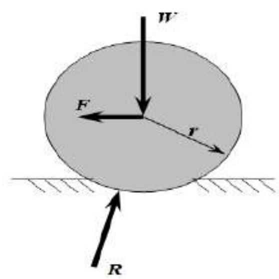
25. What is impending motion?



The state of motion of a body which is just about to move or slide is called as impending motion. When the maximum frictional force is attained and if the applied force exceeds the limiting friction, the body starts sliding or rolling. This state is called as impending motion.

26. What is rolling friction?

It is the friction experienced by a body when it rolls over another body. is the force resisting the motion when a body (such as a ball, tire, or wheel) rolls on a surface. It is mainly caused by non-elastic effects; that is, not all the energy needed for deformation (or movement) of the wheel, roadbed, etc



27. State the laws of solid friction.

The friction which exists between two surfaces that are not lubricated is called as solid friction. The two Surfaces can be at rest or one of the surfaces is moving and other surface is at rest. The following are laws of solid friction.

a. The force of friction acts in opposite direction in which the surface is having tendency to move.
b. The force of friction is equal to force applied to surfaces, so long as surface is at rest.
c. When surface is on point of motion, force of friction is highest and this maximum frictional force is called as limiting friction force.
d. The ratio between the limiting friction and normal reaction is a bit less when the two surfaces are in motion.
e. The force of friction is independent of velocity of sliding.
f. The above laws of solid friction are called as laws of static and dynamic friction or law of friction.

As the force P increases, friction F also increases but the bogy remains at rest and is in equilibrium. If F reaches a limiting value friction or from when P is increased it loses its balance and hence the body slides to right.

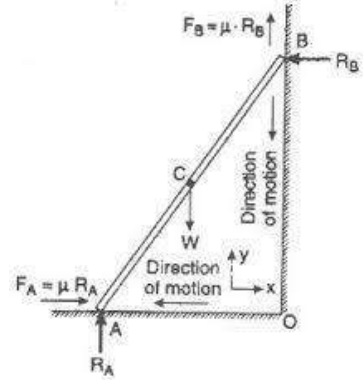
28. What is wedge?

• A wedge is a place of metal or wood in the shape of a prism whose cross section is usually a triangle or trapezoid. It is used for lifting heavy loads and tightening fits.

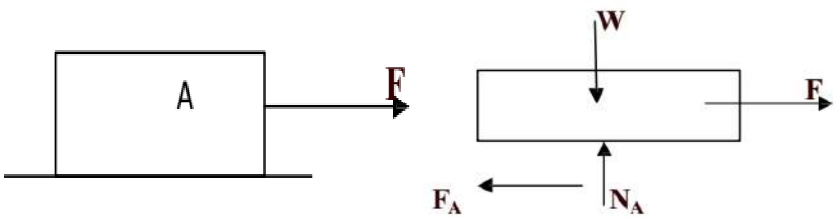
- It can be used to separate two objects or portions of an object, lift up an object, or hold an object in place. It functions by converting a force applied to its blunt end into forces perpendicular (normal) to its inclined surfaces

29. What is ladder?

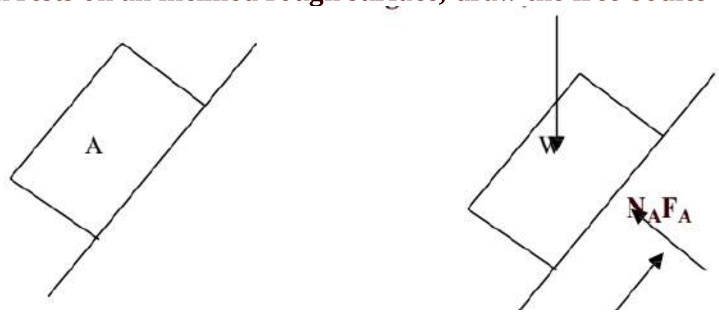
Ladder is a device for climbing or scaling on the roofs or walls. It is made by wood, iron or rope connected by number of cross pieces called ladder. The friction occur in a ladder is



30. A block of weight W, kept on a levelled rough surface is acted upon by a tensile force F. Draw the free body diagram.



31. If the above block rests on an inclined rough surface, draw the free bodies diagram



UNIT- III PROPERTIES OF SURFACES

1. **Define centre of gravity.**

   Centre of gravity of a body is the point through which the whole weight of the acts. A body is having only one centre of gravity for all position of the body. It is represented by C.G. or simply G. Centre of gravity coincides with the centre of mass if the gravitational forces are taken to be uniform and parallel. The point at which the entire weight of a body may be considered as concentrated so that if supported at this point the body would remain in equilibrium in any position.

2. **Define line of symmetry.**

   The axis about which similar configuration exist with respect to shape, size and weight on either side is known as axis of symmetry. It may be horizontal, vertical or inclined

3. **State pappus-guldinus theorem for finding surface area.**

   Theorem I:

   The area of a surface of revolution is equal to the product of length of the generating curve and the distance travelled by the centroid of the curve while the surface is being generated.

   Theorem II:

   The volume of a body or revolution is equal to the product of the generating area and the distance travelled by the centroid of the area while the body is being generated.

4. **State parallel axis theorem.**

   It states that, " if the moment of inertia of a plane area about an axis through its centre of gravity be denoted by IG, then the moment of inertia of area about axis AB parallel to the first and at a distance 'h' from the centre of gravity is given by

   $$\mathbf{I_{AB}=I_G+Ah^2}$$

   It can be used to determine the mass moment of inertia or the second moment of area of a rigid body about any axis, given the body's moment of inertia about a parallel axis through the object's center of mass and the perpendicular distance between the axes.

5. **Define axis of revolution.**

   The fixed axis about which a plane curve (may be of an arc, straight line etc.,) or a plane area is rotated is known as axis of revolution

6. **Write the expressions to find centroid of a composite plane figure.**

   X = sum of first moment of the area about y axis/Total area

   $= (a_1x_1 \pm a_2x_2) /(a_1 \pm a_2)$

   y = sum of first moment of the area about y axis/Total area

   $= (a_1y_1 \pm a_2y_2) /(a_1 \pm a_2)$

7. **State the methods of determining the centre of gravity.**

   1. By Geometrical considerations
   2. Graphical method
   3. Integration method
   4. Method of moments

8. **Define polar moment of inertia.**

   The second moment of area about a pole '0' is called the polar moment of inertia (I_p).

   $$\mathbf{I_p= I_{xx}+ I_{yy}}$$

9. **State perpendicular axis theorem.**

   It states that moment of inertia of a plane lamina about an axis perpendicular to the lamina and passing through its centroid is equal to the sum of the moment of inertias of the lamina about two mutually perpendicular axes passing through the centroid and in the plane of lamina of area 'A' with xx, yy and zz axes passing through the centroid G.

   $$\mathbf{I_{zz}=I_{xx}+I_{yy}}$$

10. **Define mass moment of inertia.**

The mass moment of inertia is one measure of the distribution of the mass of an object relative to a given axis. The mass moment of inertia is denoted by I and is given for a single particle of mass m. When calculating the mass moment of inertia for a rigid body, one thinks of the body as a sum of particles, each having a mass of dm. Integration is used to sum the moment of inertia of each dm to get the mass moment of inertia of body.

11. What is section modulus?

Section modulus is a geometric property for a given cross-section used in the design of beams or flexural members. Though generally section modulus is calculated for the extreme tensile or compressive fibres in a bending beam, often compression is the most critical case due to onset of flexural torsional buckling. I is the second moment of area (or moment of inertia) and y is the distance from the neutral axis to any given fibre.

Section modulus, z = I / Y

12. What is radius of gyration?

Radius of gyration k is the distance from the axis of rotation to the point where the entire area may be assumed to be concentrated. Radius of gyration or gyradius refers to the distribution of the components of an object around an axis. Most commonly it amounts to the perpendicular distance from the axis of rotation to the centre of mass of a rotating body. The nature of the object does not notionally affect the concept, which applies equally to a surface, a bulk mass, or an ensemble of points.

$$k = \sqrt{(I/A)}$$

13. What is inertia of body?

It is the property by which the body resists any change in its state of rest or of uniform motion. Inertia is the resistance of any physical object to any change in its state of motion, including changes to its speed and direction. It is the tendency of objects to keep moving in a straight line sat constant velocity. The principle of inertia is one of the fundamental principles of classical physics that are used to describe the motion of objects and how they are affected by applied forces.

14. Define moment of inertia

Moment of inertia of a body about any axis is the property by which the body resists rotation about that axis. It is denoted by letters M.I. It is the mass property of a rigid body that determines the torque needed for a desired angular acceleration about an axis of rotation. Moment of inertia depends on the shape of the body and may be different around different axes of rotation. A larger moment of inertia around a given axis requires more torque to increase the rotation, or to stop the rotation, of a body about that axis. Moment of inertia may also be called mass moment of inertia, rotational inertia, polar moment of inertia, or the angular mass.

15. Write the moment of inertia formula for rectangular section.

Let Ixx and Iyy be the moment of inertia of the rectangular block about the xx axis and yy axis.

$I_{xx} = bd^3/12$

$I_{yy} = db^3/12$

Where, b- breadth and d- depth

16. Write the moment of inertia formula for triangular section.

Let Ixx and Iyy be the moment of inertia of the triangular section about the xx axis and yy axis.

Moment of inertia about the base (AB)

$I_{AB} = bh^3/12$

Moment of inertia about centroidal axis

$I_{xx} = bh^3/36$

Where, b- breadth and h- height

17. Write the moment of inertia formula for circular section.

Let Ixx and Iyy be the moment of inertia of the circular section about the xx and yy axis.

$I_{yy}=I_{xx}= \pi r^4 /4$

Where, r-radius of circle

18. Write the moment of inertia formula for semi-circular section.

Let Ixx and Iyy be the moment of inertia of the block about the xx and yy axis.

$I_{xx}= 0.11\ r4$

$I_{yy}= \pi d^4 /128$

Where, d and r are the radius and diameter of circle.

19. What is meant by product of inertia?

Relative to two rectangular axes, the sum of the products formed by multiplying the mass (or, sometimes, the area) of each element of a figure by the product of the coordinates corresponding to those axes. The values of the products of inertia depend on the orientations of the coordinate axes. For every point of the body or system, there exist at least three mutually perpendicular axes, called the principal axes of inertia, for which the products of inertia are equal to zero.

20. Differentiate mass moment of inertia and area moment of inertia.

Area moment of inertia	**Mass moment of inertia**
Masses are assumed to be concentrated and hence area of plane figures are taken to find the moment of inertia	It is consider only for solid figure
Radius of gyration of plane figure about any axis is defined as the distance at which the entire area A of the figure is assumed to be concentrated, such that the moment of inertia of area A about the axis is same	Radius of gyration of solid figure about any axis is defined as the distance at which the entire mass M of the figure is assumed to be concentrated, such that the moment of inertia of mass M about the axis is same

21. What is composite section?

Composite Construction of Buildings refers to any members composed of more than one material. The parts of these composite members are rigidly connected such that no relative movement can occur. Composite beams are normally hot rolled or fabricated steel sections that act compositely with the slab. The composite interaction is achieved by the attachment of shear connectors to the top flange of the beam.

UNIT IV – KINEMATICS AND KINETICS OF PARTICLES

1. **What is projectile?**
   A particle, moving under the combined effect of vertical and horizontal forces is called a projectile
2. **Distinguish between kinetics and kinematics.**
   Kinetics and kinematics are two words in the study of motion and the forces that are involved in these motions that confuse a lot of people. The situation becomes confusing because these two words are similar sounding, and also because both of them are involved in the study of motion. However, while kinetics study the motion and the forces that are underlying this motion, kinematics is solely focused on the study of motion and does not take into account any forces that may be acting upon the body in motion.
3. **State the law of conservation of momentum.**
   It states that, if the resultant force acting on a particle is zero, then the linear momentum of the particle remains constant ie, Final momentum = Initial momentum.
4. **State newton's law of collision of elastic bodies.**
   It states that "for two colliding bodies, their relative velocity of separation bears a constant ratio to their relative velocity of approach"
5. **Write the principle of work and energy.**
   It states that "when a particle moves from position, S1 to S2 under the action of a force F, the change in kinetic energy of the particle is equal to the force F"
6. **What is conservative force?**
   A force F is said to be conservative, when the force components are derivable from a potential and the work done by the force F between any two points is independent of the path followed
7. **State theorem of conservation of energy.**
   Law of conservation of energy- the **law of conservation of energy** states that the total energy of an isolated system remains constant—it is said to be *conserved* over time. Energy can be neither created nor be destroyed, but it can change form, for instance chemical energy can be converted to kinetic energy in the explosion of a stick of dynamite.
   A consequence of the law of conservation of energy is that a perpetual motion machine of the first kind cannot exist. That is to say, no system without an external energy supply can deliver an unlimited amount of energy to its surroundings
8. **Write impulse-momentum equation.**
   F.t =m (v- u)
   F- Impulsive force
   m- Mass of the body
   v- Final velocity
   u- Initial velocity
9. **What is line of impact?**
   It is an imaginary line passing through the point of contact and normal to the plane of contact.
10. **Define angle of friction.**
    Angle of friction is the angle between the line of action of the total reaction of one body on another and the normal to the common tangent between the' bodies when motion is impending.

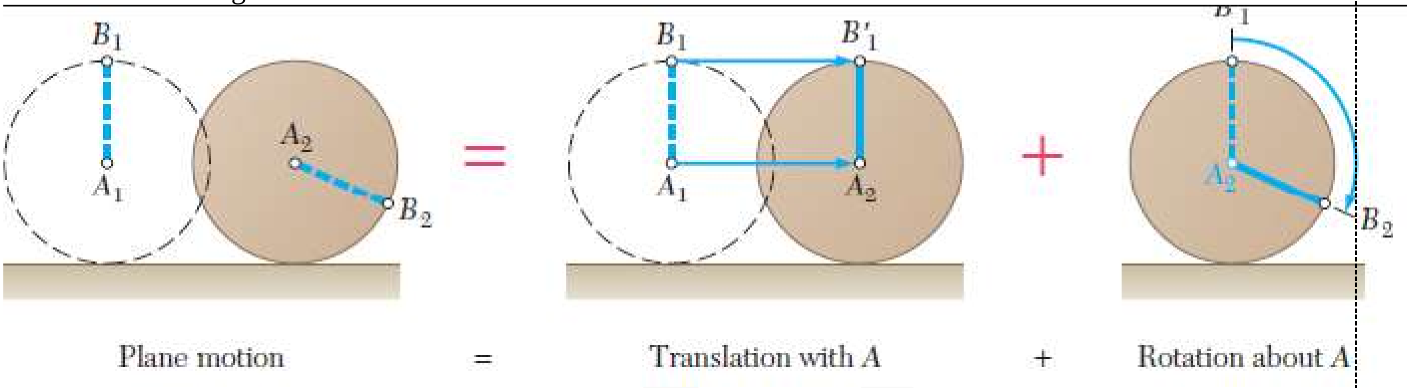
UNIT V – KINEMATICS AND KINETICS OF RIGID BODIES

1. **Define instantaneous centre of rotation.**

In general plane motion of a rigid body at any given instant the velocities of various particles of the slab are same as if the slab bear rotating about certain axis perpendicular to the plane of the slab. Instantaneous centre of rotation is a point identified with in a body where the velocity is zero.

2. **What is general plane motion ?give some examples.**

If a rigid body moves with both translational and rotational motion, it is said to be in general plane motion. General Plane Motion Force and Acceleration Diagrams around the Center of Gravity is shown in figure.



3. **Define Angular momentum.**

Angular momentum-, **moment of momentum**, **or rotational momentum** is a measure of the amount of rotation an object has, taking into account its mass, shape and speed. It is a vector quantity that represents the product of a body's rotational inertia and rotational velocity about a particular axis. The angular momentum of a system of particles (e.g. a rigid body) is the sum of angular momenta of the individual particles. For a rigid body the angular momentum can be expressed as the product of the body's moment of inertia, *I*, (i.e., a measure of an object's resistance to changes in its rotation velocity) and its angular velocity, ω.

4. **What are motion curves?**

A motion curve is just one of the animated parameters, considered as a function of time. For example, if p1(t) is indeed our function mapping from time to the x-coordinate of the camera in a scene, we can plot it to see it as a curve. If we felt something was wrong with the camera motion, looking at that plot could instantly clue us in to moments where it might be too jerky or overly smooth.

5. **What is relation b/w angular velocity and linear velocity?**

Sl.No	Linear Velocity	Angular Velocity
1	It is the velocity of object travelling in the **straight line**.	It is the velocity of object traveling in the **circular or curved or angular path.**
2	It is measured in **m/sec.**	It is measured in **rad/sec.**
3	The equation of Linear velocity is: v = *xt***m/sec**	The equation of Angular velocity is: *ω* = *θt***radians/sec**
4	The linear velocity could be constant since the object could be travelling with constant speed without changing its direction.	The angular velocity could not be constant since the body is constantly changing its direction.

6. **Define projectile motion.**

Projectile motion is a form of motion in which an object or particle (called a projectile) is thrown near the earth's surface, and it moves along a curved path under the action of gravity only. The only force of significance that acts on the object is gravity, which acts downward to cause a downward acceleration. In projectile motion, the horizontal motion and the vertical motion are independent of each other; that is, neither motion affects the other.

7. **Define time of flight.**

Time of flight (TOF) describes a variety of methods that measure the time that it takes for an object, particle or acoustic, electromagnetic or other wave to travel a distance through a medium. This measurement can be used for a time standard (such as an atomic fountain), as a way to measure velocity or path length through a given medium, or as a way to learn about the particle or medium (such as composition or flow rate). The traveling object may be detected directly (e.g., ion detector in mass spectrometry) or indirectly (e.g., light scattered from an object in laser Doppler velocimetry).

8. Write short notes on potential Energy.

An object can store energy as the result of its position. For example, the heavy ball of a demolition machine is storing energy when it is held at an elevated position. This stored energy of position is referred to as potential energy. Similarly, a drawn bow is able to store energy as the result of its position. When assuming its *usual position* (i.e., when not drawn), there is no energy stored in the bow. Yet when its position is altered from its usual equilibrium position, the bow is able to store energy by virtue of its position. This stored energy of position is referred to as potential energy. **Potential energy** is the stored energy of position possessed by an object.

9. Write short notes on power.

The expression for power is work/time. And since the expression for work is force x displacement, the expression for power can be rewritten as (force*displacement)/time. Since the expression for velocity is displacement/time, the expression for power can be rewritten once more as force x velocity. This is shown below.

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{\text{Force} \cdot \text{Displacement}}{\text{Time}}$$

$$\text{Power} = \text{Force} \cdot \frac{\text{Displacement}}{\text{Time}}$$

$$\boxed{\text{Power} = \text{Force} \cdot \text{Velocity}}$$

10. What is time of restitution, compression and collision?

i. The time taken by the two bodies to regain original shape, after compression is known as time of restitution.

ii. The time taken by the two bodies in compression, after the instant of collision is known as time of compression.

iii. The sum of time of compression and time of restitution is known as time of collision or period of collision or period of impact.

11. Write short notes on collision.

A collision is an event in which two or more bodies exert forces on each other for a relatively short time. Although the most common colloquial use of the word "collision" refers to accidents in which two or more objects collide, the scientific use of the word "collision" implies nothing about the magnitude of the forces.

Some examples of physical interactions that scientists would consider collisions:

i. An insect touches its antenna to the leaf of a plant. The antenna is said to collide with leaf.

ii. A cat walks delicately through the grass. Each contact that its paws make with the ground is a collision. Each brush of its fur against a blade of grass is a collision. Write short notes on Impact while collision of two bodies.